

Nitheesh Karthikeyan

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ECE student with hands-on experience in automation, systems development, and data-driven engineering — eager to apply technical foundations to ADAS software development and testing in automotive environments.

EDUCATION

VELLORE INSTITUTE OF TECHNOLOGY

B.Tech in Electronics and Communication Engineering

2024--2028 | Chennai, India
Coursework relevant to embedded systems, signals, communication protocols, and digital design.

SBOA SCHOOL AND JR. COLLEGE

2010--2024 | Chennai, India
Winner in **Model United Nations**; strong structured communication and analytical skills.

T-SHAPED SKILL SET

Programming/HDL Languages:
C++, Python, C, Java, VHDL, Verilog, Assembly

Embedded & Systems:
ECU fundamentals, Communication Protocols (CAN, LIN), Linux (working knowledge)

ADAS & Sensor Systems:
Ultrasonic transducer design & simulation, Sensor array prototyping, Camera / Radar / Lidar , ADAS fundamentals

Test & Validation:
Test automation (pipeline design & execution), Automated workflow validation, HiL/MiL/SiL, CI/CD concepts

Web/Frontend (Secondary):
Next.js, React

LANGUAGES

English — Fluent
Tamil — Native
Hindi — Professional Working
French — Limited Proficiency

LEADERSHIP

Zerodha
Zerodha Varsity Lead Coordinator
Managed a cross-functional 100-member team for smooth execution of India’s premier financial literacy competition with 500+ participants and INR 3.5 Lakhs+ Prize Pool. Maintained detailed accountability frameworks across teams, ensuring quality-first execution. Re-invited for two subsequent engagements outside the university based on prior conduct of leadership qualities.

CERTIFICATIONS & VIRTUAL EXPERIENCE

Siemens Mobility — Commercial PM simulation (Forage): analyzed \$100M+ transport project using WBS, KPIs, and EAC; gained exposure to industrial project lifecycle and stakeholder reporting.

EXPERIENCE

MEDXAI RESEARCH SOFTWARE INTERN (WITH STIPEND)

October 2024 -- February 2026

- Conducted research on **flexible array transducers** for ultrasound imaging; performed **structural and acoustic simulations in SolidWorks and Blender** for biomedical clients developing a flexible array ultrasound (USG) probe — progressing from simulation to a **working ultrasound scanner prototype**. Demonstrates end-to-end sensor development experience directly analogous to **ADAS sensor modelling and validation pipelines**.
- Contributed to biomedical MRI and ultrasound hardware design cycles; translated simulation outputs into design decisions for physical transducer arrays — experience closely paralleling **sensor characterization workflows for Camera, Radar, and Ultrasonic ADAS systems**.
- Automated end-to-end **computational workflows** for drug discovery across **20+ pharmaceutical clients**, reducing manual effort and narrowing viable candidate search space — directly analogous to **test pipeline automation** in embedded development contexts.
- Contributed to **one drug candidate passing wet-lab validation** based on computational predictions; demonstrated accuracy and reliability of automated systems under real-world constraints.
- Engineered **5 internal tools** including **Drug Explorer**, a data ingestion and visualization pipeline converting SMILES inputs into structured decision-support insights; now **outsourced to major pharmaceutical companies**.
- Operated within **Agile/Scrum environments** — sprint planning, stand-ups, and iterative release cycles using **Jira**; collaborated across cross-functional global teams.
- Produced structured **technical reports and dashboards** for multinational clients (e.g., OmniActive), translating complex data outputs into actionable engineering decisions.

PROJECTS

- ML-Based Library Seat Occupancy System** — Developed and deployed a **real-time sensor-driven system** using computer vision and ML to track and visualize seat occupancy; actively used in the campus library. Awarded the **Vice President’s Feather-in-the-Cap Award, VIT**. Demonstrates end-to-end system development from sensor input to live data visualization.
- LithoMap** — Clinical Research Data Visualization Platform. Built a **production-grade application used daily by researchers** at Government Stanley Medical College to encode and visualize FTIR clinical data into structured records. Identified workflow bottlenecks, designed a data pipeline solution, and iterated with domain experts for real-world deployment — paralleling embedded HW/SW integration cycles.
- Financial Fraud Detection System** (Qinnovision 2025, NYU Abu Dhabi) — Privacy-first anomaly detection system using **quantum federated ML** to identify fraudulent transactions; designed for accuracy, scalability, and robustness under distributed constraints.
- Pothole Detection System** — Computer vision-based road anomaly detection system; recognized among **Top 3 Non-Startup Projects at VIT Chennai 2024** (iInventiv, IIT Madras). Real-world perception task analogous to ADAS object detection pipelines.

HACKATHONS

- IBM Qiskit Fall Fest 2024 — 1st Place Winner**. Presented to senior leadership (Dr. L. Venkataraman) at IBM Quantum India; reviewed by international faculty.
- IBM Qiskit Fall Fest 2025 — Judge & Teaching Instructor**. Delivered quantum programming sessions to **100+ international participants** from 10+ countries; facilitated live technical discussions and mentoring.
- Intel + Microsoft SAS Hackathon** (HEC Montréal) — **World Rank 19**. Optimized business outcomes by analyzing user flow data and applying mathematical models to maximize system performance — strong emphasis on data-driven decision-making.
- ISRO Analog Circuits Hackathon, IIST Trivandrum - Rank 5** Collaborated on designing analog front-end circuits using operational amplifiers for sensor data acquisition and filtering. Leveraged basic op-amp configurations (inverting/non-inverting amplifiers, comparators, active filters) to implement signal conditioning stages that improved measurement accuracy.

AWARDS

- United Nations Millennium Fellow (2025)** — Top **4%** of 60,000+ applicants; selected for global academic and social impact through technology.
- 1st Place – Best Startup Idea Award (2025)** — Presented by Dr. Soumya Swaminathan, former Chief Scientist, World Health Organization.
- Selected as **1 of 5 students** from VIT Chennai for the **Nokia Campus Connect Program** — rigorous training in communication systems, machine learning, and computer science.
- Vice President’s Feather-in-the-Cap Award, VIT** — For developing and deploying an **ML-based Seat Occupancy System** in the VIT Library.
- Recognized among **Top 3 Non-Startup Projects at VIT Chennai 2024** for a Pothole Detection System (IIT Madras, iInventiv).
- University Day 2026** Award for developing a Map Application for international students and visiting professors.